

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – CONCEPT MAPPING

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 2 – Concept Mapping
Weightage:	20% of CIA	Total Marks:	100
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:		Reg. No.:	
Section / Batch:		Date:	

Instructions to Students

- Answer the following question. Total Marks: 100.
- Draw a clear and neat concept map for each question.
- Identify all key concepts and arrange them in a logical hierarchy.
- Show meaningful linkages between concepts using arrows and connecting words.
- Compare related concepts wherever required and justify the relationships clearly.
- Ensure the concept map is well-organized, readable, and properly labelled.

Question Type	Focus Area	Questions
Concept Mapping	Map algorithm efficiency concepts using asymptotic analysis, search techniques, recurrence relations, and recursion tree method	Q1

SECTION A – CONCEPT MAPPING

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%				
Linkages	25%				
Organization	20%				
Integration of Literature	20%				
Clarity	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name: _____	Name: _____	Name: _____
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

SPACE COMPLEXITY

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The amount of memory space an algorithm requires as a function of input size (n). It includes both fixed and variable parts.

1. COMPONENTS

a) Fixed Space	b) Variable Space
Space used by the program instructions Does not depend on input size	Space used by data structures, recursion stack, dynamic memory etc
constant amount	Depends on input size (n)
Eg: variables, constants, instructions	Eg: Arrays, linked lists, recursion calls

2. TYPES OF SPACE COMPLEXITY

a) Auxiliary Space	b) Total Space
* Extra Space used by the algorithm apart from input	* Sum of space used for input and auxiliary and auxiliary space
* Does not include space for input data	* Total space = Input space + Auxiliary spaces
Eg: Temporary variables, recursion stack, extra data structures	Eg: If input array needs n space and extra space is n , total space = $2n$

3. COMMON COMPLEXITIES (Asymptotic Notation)

- $O(1)$ - constant space
- $O(\log n)$ - logarithmic space
- $O(n)$ - linear space
- $O(n \log n)$ - linearithmic space
- $O(n^2)$ - Quadratic space
- $O(2^n)$ - Exponential space

4. SPACE COMPLEXITY IN DIFFERENT SCENARIOS

a) Iterative	b) Recursive	c) Data structures
usually use constant or linear space	use stack space for function calls	different data structures require different space
Eg: Linear Search uses $O(1)$ space	Eg: Factorial (n) uses $O(n)$ spaces	Eg: Array - $O(n)$ Linked list - $O(n)$

5. ANALYSIS APPROACH

- Step 1: Identify all memory used (instructions, variable, data structures, function calls)
- Step 2: Express using asymptotic notation (O, Θ, Ω)

6. IMPORTANCE

- Helps in choosing memory-efficient algorithms
- Important for large input size
- crucial in embedded systems, mobile devices and applications with limited memory